Mathematical Model Building

Basic Financial Calculations

$f(x)$

$f(x)$

Finance Terminology and Calculations
Basic business example

You consider opening a small retail store for 3 years and see if it works out. Starting out requires an initial investment of $1,000.

- I_0 = initial investment ($1,000)

Each year it costs $10 to operate.

- C = annual operating costs ($10)

Each year it makes sales of $100.

- R = annual revenue ($100)

$f(x)$

Finance Terminology and Calculations
Cash flow

Annual cash flow is the net amount of money flowing in and out in the year

- CF = annual cash flow
- $CF = R - C$

Cash flows must be specified by period because, unlike our simple example, they are generally different from period to period

- t = index for the period (month, quarter, year, etc.)
- $CF_t = R_t - C_t$
- $CF_0 = -I_0$ *cash flow in year $t = 0$ is the initial investment I_0

Finance Terminology and Calculations
Cumulative Cash flow

The cumulative cash flow is the year over year sum of annual cash flows

- $CCF_t =$ cumulative cash flow up through period t
- $CCF_t = \sum_{i=0}^{t} CF_i$
- $CCF_t = -I_0 + \sum_{i=1}^{t} (R_i - C_i)$ * This is also easier to setup in Excel

$f(x)$

Finance Terminology and Calculations
Cash flows over time

With an initial investment of $1,000, annual operating costs of $10, and annual revenue of $100.

These are the undiscounted cash flows for 3 years.

Year t	$t = 0$	$t = 1$	$t = 2$	$t = 3$
R_t: Revenue	—	$100	$100	$100
C_t: Operating costs	—	($10)	($10)	($10)
I_0: Initial investment	($1,000)	—	—	—
CF_t: Net Cash Flow in year t	($1,000)	$90	$90	$90
CCF_t: Cumulative Cash Flow thru year t	($1,000)	($910)	($820)	($730)

$f(x)$

Finance Terminology and Calculations
Cash flows with varying costs and revenues

Assume operating costs increase by $10 per year, and annual revenue increases by $500 per year.

These are the revised, undiscounted cash flows for 3 years.

Year t	$t = 0$	$t = 1$	$t = 2$	$t = 3$
R_t: Revenue	—	$100	$600	$1,100
C_t: Operating costs	—	($10)	($20)	($30)
I_0: Initial investment	($1,000)	—	—	—
CF_t: Net Cash Flow in year t	($1,000)	$90	$580	$1,070
CCF_t: Cumulative Cash Flow thru year t	($1,000)	($910)	($330)	$740

Time Value of Money

$f(x)$

$f(x)$

Time Value of Money
A dollar today is worth more than a dollar next year.

Assume you can earn a 10% return over one year on $100 invested today.

Then $100 invested today will be worth $110 one year from now.

$$\$100 * (1 + 0.10) = \$110$$

- r = rate of return (10%)
- PV = present value ($100)
- t = number of periods the present value earns the rate of return r
- FV = what the amount invested today will be worth in t periods

$$FV = PV * (1 + r)^t$$

$$\$110 = \$100 * (1 + 0.10)^1$$

Time Value of Money
Excel Future Value Formula - lump sum

FV = what the amount invested today will be worth in t periods

=fv (rate, nper, pmt, [pv], [type])

For a single lump sum amount

- rate = interest rate per period (r)
- nper = number of payment periods (t)
- pv = present value = (PV)

$$FV = PV * (1 + r)^t$$

= fv(r, t, pmt, PV, [type])

= fv(0.10, 1, , -100,)

= 110

- Excel's cash outflow is negative – invest $100 today
- Excel's cash inflow is positive – receive $110 next year

$f(x)$

Time Value of Money
What is a dollar next year worth today

Assume you will inherit $100 next year and that the current rate of return, or **discount rate**, is 10%.

What will $100 a year from now be worth in today's dollars?

- r = discount rate (10%)
- t = number of periods the future value will be discounted at the rate r
- FV = the amount received in t periods from today ($100)
- PV = what the FV in t periods is worth today with a discount rate of r

$FV = PV * (1 + r)^t$ solving for PV gives $PV = \frac{FV}{(1+r)^t}$

$$PV = \frac{FV}{(1+r)^t} = \frac{100}{(1+0.1)^1} = \$90.91$$

Time Value of Money
Excel Present Value Formula - lump sum

PV = what the FV in t periods is worth today with a discount rate of r

=pv (Rate, Nper, Pmt, Fv, [type])

For a single lump sum amount

- Rate = discount rate (r)
- Nper = number of payment periods (t)
- Fv = future value = (FV)

$$PV = \frac{FV}{(1+r)^t}$$

= pv(r, t, pmt, FV, [type])

= pv(0.1, 1, , -100,)

=90.91

- Excel's cash inflow is positive so PV = $90.91 today
- Excel requires that PV and FV have opposite signs so FV = -100

Net Present Value (NPV)

$f(x)$

$f(x)$

Time Value of Money
Net Present Value

NPV = the **value today** of future cash flows minus the initial investment.

Recall that cumulative cash flow is the sum of the cash flows over time, including the initial investment

$$CCF_t = -I_0 + \sum_{i=1}^{t} (R_i - C_i) \qquad CF_t = R_t - C_t$$

The value of a future cash flow today is:

$$PV = \frac{FV}{(1+r)^t}$$

The value today of all the future cash flows requires they be discounted

$$NPV = -I_0 + \sum_{i=1}^{t} \frac{CF_i}{(1+r)^i}$$

Time Value of Money
Net Present Value - example

Year t	$t = 0$	$t = 1$	$t = 2$	$t = 3$
R_t: Revenue	—	$100	$100	$100
C_t: Operating costs	—	($10)	($10)	($10)
I_0: Initial investment	($1,000)	—	—	—
CF_t: Net Cash Flow in year t	($1,000)	$90	$90	$90
CCF_t: Cumulative Cash Flow thru year t	($1,000)	($910)	($820)	($730)

$$NPV = -I_0 + \sum_{i=1}^{t} \frac{CF_i}{(1+r)^i}$$

$$= \quad -1000 + \frac{90}{1.1} + \frac{90}{1.1^2} + \frac{90}{1.1^3} = -\$776$$